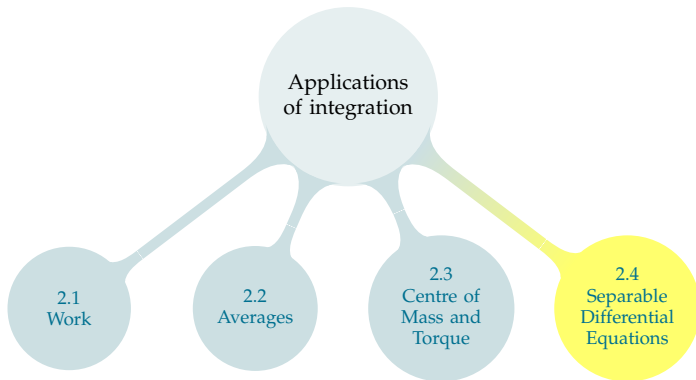


TABLE OF CONTENTS



Differential Equation

A **differential equation** is an equation for an unknown function that involves the derivative of the unknown function.

Differential equations play a central role in modelling a huge number of different phenomena. Here is a table giving a bunch of named differential equations and what they are used for. It is far from complete.

Newton's Law of Motion	describes motion of particles
Maxwell's equations	describes electromagnetic radiation
Navier-Stokes equations	describes fluid motion
Heat equation	describes heat flow
Wave equation	describes wave motion
Schrödinger equation	describes atoms, molecules and crystals
Stress-strain equations	describes elastic materials
Black-Scholes models	used for pricing financial options
Predator-prey equations	describes ecosystem populations
Einstein's equations	connects gravity and geometry
Ludwig-Jones-Holling's equation	models spruce budworm/Balsam fir ecosystem
Zeeman's model	models heart beats and nerve impulses
Sherman-Rinzel-Keizer model	for electrical activity in Pancreatic β -cells
Hodgkin-Huxley equations	models nerve action potentials

Disclaimer:

We are dipping our toes into a vast topic. Most universities offer half a dozen different undergraduate courses on various aspects of differential equations. We will just look at one special, but important, type of equation.

- ▶ We will first learn to **verify** solutions without **finding** them. (If you learned about differential equations last semester, this will be review.)
- ▶ **Then**, we will learn to solve one particular type of differential equation.

DIFFERENTIAL EQUATIONS

Definition

A **differential equation** is an equation involving the derivative of an unknown function.

Examples: $\frac{dy}{dx} = 2x$; $x\frac{dy}{dx} = 7xy + y$

Definition

If a **function** makes a differential equation true, we say it **satisfies** the differential equation, or is a solution to the differential equation.

Example: $y = x^2$ and $y = x^2 + 1$ both satisfy the first differential equation

VERIFYING SOLUTIONS

Consider the equation

$$x + 2 = x^3 - x^2$$

How would you verify whether $x = 1$ satisfies the equation?

How would you verify whether $x = 2$ satisfies the equation?

Plug x into the equation, check whether the left-hand side and the right-hand side are the same **number**.

VERIFYING SOLUTIONS

Consider the differential equation

$$\frac{dy}{dx} = 2y + 4x$$

How would you verify whether $y = e^{2x} - 2x$ satisfies the equation?

How would you verify whether $y = e^{2x} - 2x - 1$ satisfies the equation?

Replace y and $\frac{dy}{dx}$ in the equation, check whether the left-hand side and the right-hand side are the same **function**.

Differential equation:

$$x \frac{dy}{dx} = 7xy + y$$

Interpretation:

There is a function $y(x)$ that makes the left-hand side and the right-hand side into the same function.

To check whether a given function satisfies the differential equation, plug it in for everything with a “ y ”: y itself and $\frac{dy}{dx}$.

Is $y = xe^{7x+9}$ a solution to the differential equation?

Which of the following solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$?

A. $y = -x$

B. $y = x + 5$

C. $y = \sqrt{x^2 + 5}$

FIRST EXAMPLE OF A SEPARABLE DE

Definition

A separable differential equation is an equation for a function $y(x)$ that can be written in the form

$$g(y) \cdot \frac{dy}{dx} = f(x)$$

(It may take some rearranging to get the equation into this form.)

For example:

$$y^2 \cdot \frac{dy}{dx} = 4x$$

GENERAL METHOD FOR SOLVING SEPARABLE DES

$$g(y) \cdot \frac{dy}{dx} = f(x)$$

GENERAL METHOD FOR SOLVING SEPARABLE DES

$$g(y) \cdot \frac{dy}{dx} = f(x)$$

Shorthand:

$$g(y) \cdot \frac{dy}{dx} = f(x)$$

$$g(y) \, dy = f(x) \, dx$$

$$\int g(y) \, dy = \int f(x) \, dx$$

$$\frac{dy}{dx} = y^2 x$$

1. "Separate" y 's from x 's.
2. Integrate.
3. Solve explicitly for y .

$$\frac{dy}{dx} = (xy)^4, \quad y(0) = \frac{1}{2}$$

$$\frac{dy}{dx} = y(4x^3 - 1) \quad y(0) = -2$$



Let a and b be any two constants. We'll now solve the family of differential equations

$$\frac{dy}{dx} = a(y - b)$$

using our mnemonic device.

Linear First-Order Differential Equations

Let a and b be constants. The differentiable function $y(x)$ obeys the differential equation

$$\frac{dy}{dx} = a(y - b)$$

if and only if

$$y(x) = \{y(0) - b\} e^{ax} + b$$

Find a function $y(x)$ with $y' = 3y + 7$ and $y(2) = 5$.

The rate at which a medicine is metabolized (broken down) in the body depends on how much of it is in the bloodstream. Suppose a certain medicine is metabolized at a rate of $\frac{1}{10}A \mu\text{g/hr}$, where A is the amount of medicine in the patient. The medicine is being administered to the patient at a constant rate of $2 \mu\text{g/hr}$. If the patient starts with no medicine in their blood at $t = 0$, give the formula for the amount of medicine in the patient at time t . What happens to the amount over time?

Linear First-Order Differential Equations

Let a and b be constants. The differentiable function $y(x)$ obeys the differential equation

$$\frac{dy}{dx} = a(y - b)$$

if and only if

$$y(x) = \{y(0) - b\} e^{ax} + b$$

$$\frac{dA}{dt} = 2 - \frac{1}{10}A = -\frac{1}{10}(A - 20) \quad A(0) = 0$$